Lets have some fun

Agenda

- Learn about approximations
- Example of approximation
- Homework

Introducing Analysis

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- Building the basic number rings

But first… what is a set?

- A number of distinct objects that form a **group**.

First numbers

- Natural numbers $\mathbb{N}$
- 1,2,3,4,5,6,7,8,9
- 0 is included depending on the school

Second Ring

- Negative naturals and positive naturals
- $-(\mathbb{N}) \cup \mathbb{N} \cup \{0\}$
- Integers $\mathbb{Z}$

Third Ring

- Lets add fractional values
- $\mathbb{Q} = \{\frac{p}{q} \,|\, p, q \in \mathbb{Z}, q \neq 0\}$
- Rational numbers $\mathbb{Q}$

Questions?

- Does Q account for all numbers?

Irrational numbers

- Create these by approximation
- $\pi, e, \sqrt{2}, \ldots$
- irrationals $\mathbb{P}$

Real numbers

- Union rational and irrational numbers
- $\mathbb{R} = \mathbb{Q} \cup \mathbb{P}$

Homework